



SARS CoV-2 Dispatches

Efflux pump gene single-nucleotide variants associated with resistance in *Mycobacterium tuberculosis* isolates with discrepant drug genotypesZahra Hasan^{*,*}, Safina Abdul Razzak[#], Akbar Kanji, Sadia Shakoar, Rumina Hasan

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ABSTRACT

Introduction: Single-nucleotide variants (SNVs) in *Mycobacterium tuberculosis* (*M. tuberculosis*) genomes can predict multidrug resistance (MDR) but not all phenotype-genotype correlations can be explained. We investigated SNVs in efflux pumps (EPs) in the context of *M. tuberculosis* drug resistance.

Methods: We analysed 2221 *M. tuberculosis* genomes from 1432 susceptible and 200 MDR, 172 pre-extensively drug resistant (XDR) and 417 XDR isolates. Analysis of 47 EP genes was conducted using MTB-VCF, an in-house bioinformatics pipeline. SNVs were categorized according to their SIFT/Polyphen scores. Resistance genotypes were also called using the TB-Profiler tool.

Results: Genome comparisons between susceptible and drug resistant (DR) isolates identified 418 unique SNVs in EP of which; 53.5% were in MDR, 68.9% in pre-XDR and 61.3% in XDR isolates. Twenty EPs had unique SNVs with a high SIFT/PolyPhen score, comprising 38 unique SNVs. Sixteen SNVs across 12 EP genes were significantly associated with drug resistance and enriched in pre-XDR and XDR strains. These comprised 12 previously reported SNVs (in Rv0191, Rv0507, Rv0676, Rv1217, Rv1218, Rv1273, Rv1458, Rv1819, and Rv2688) and 4 novel SNVs (in Rv1877 and Rv2333). We investigated their presence in genomes of 52 MDR isolates with phenotype-genotype discrepancies to rifampicin (RIF), isoniazid (INH), or fluoroquinolones. SNVs associated with RIF and INH (Rv1217_1218, Rv1819, Rv0450, Rv1458, Rv3827, Rv0507, Rv0676, Rv1273, and Rv2333), and with fluoroquinolone (Rv2688) resistance were present in these discrepant strains.

Conclusions: Considering SNVs in EPs as part of *M. tuberculosis* genome-based resistance interpretation may add value, especially in evaluation of XDR resistance in strains with phenotype-genotype discrepancies.

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1. Introduction

Tuberculosis (TB) remains a global public health concern and drug resistance adds to the complexity and cost of treatment. According to the World Health Organization (WHO), an estimated 1.3 million deaths were reported in 2020 [1]. Early identification of drug resistant (DR) *Mycobacterium tuberculosis* (MTB) is important for effective management of multi drug-resistant (MDR) TB, defined by resistance to rifampicin (RIF) and isoniazid (INH). TB is usually diagnosed by acid-fast bacillus (AFB) smear microscopy and/or Xpert *M. tuberculosis*/RIF testing. With a positive result and absence

of RIF resistance, patients are given first-line treatment with RIF, INH, pyrazinamide (PZA), and ethambutol (EMB). AFB culture for phenotypic drug susceptibility testing (DST) is performed where available. If MDR-TB or rifampicin-resistant TB (RR-TB) is diagnosed, both phenotypic and genotypic susceptibility testing for fluoroquinolones (FLQ) and aminoglycosides is indicated [2,3]. However, neither the phenotypic nor genotypic method is perfect and there can be discordance between results especially in the case of INH, FLQ, and aminoglycosides [4–6].

M. tuberculosis whole-genome sequencing (WGS) provides information on genes associated with resistance to first-line (RIF, INH, PZA, and EMB) drugs used for treatment of drug-sensitive TB, FLQ, and aminoglycosides, as well as new and repurposed drugs, such as bedaquiline (BDQ), clofazimine (CFZ), delamanid (DMN), and linezolid [7,8]. Mutations, single-nucleotide variants (SNVs) associated with drug resistance can be interpreted by bioinformatics

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tools, such as TBProfiler [8,9], PhyResSE [10], and KVARQ [11,12]. The WHO TB Mutation catalogue 2021 provides guidance on SNVs with high predictive value of resistance to anti-TB drugs [13]. Next-generation sequencing (NGS)-based interpretation also adds value by identifying resistance genotypes in the absence of DST [14]. However, *M. tuberculosis* phenotypic resistance can remain unexplained even after the investigation of known genotypes associated with resistance [15]. There is a need to understand additional mechanisms that contribute to DR in *M. tuberculosis*.

Efflux pumps (EPs) contribute to drug resistance in various bacterial species and *M. tuberculosis* has one of the largest numbers of putative EPs amongst different bacteria [16]. A number of well-characterized EPs are shown to play a role in drug resistance include, ATP binding cassette (ABC), Major Facilitator Super-family (MFS), RND family proteins, small molecule transporter (SMR), and Amino acid permease family (APC) proteins types [17–19]. The role of EP is further substantiated as pump inhibitors, such as verapamil, carbonyl cyanide m-chlorophenylhydrazone, chlorpromazine, and reserpine, reduce resistance to anti-TB drugs [20]. *In vitro* and *in vivo* studies have demonstrated the effect of EPs in drug resistance to *M. tuberculosis*. For example, overexpression of *Rv3065*, *Rv2938*, *Rv1819*, *Rv2209*, *Rv2459*, *Rv2477c*, *Rv2688*, *Rv2846*, *Rv2994*, and *Rv3728* induced by drug pressure is associated with resistance to INH, EMB, and STR [21]. Whereas expression studies require complex experimentation whereas genome sequencing of *M. tuberculosis* provides genotypic data that can be analysed in the context of EP for isolates with genotype-phenotype discrepant profiles. Importantly, EPs exert their effect separately from that of high-confidence mutations (HCMs) that are associated with drug resistance [22], therefore providing additional mechanisms of resistance even in non-discrepant strains.

Previously, SNVs in *M. tuberculosis* EP *Rv0194*, *Rv1217*, *Rv1218*, *Rv2936*, *Rv2937*, *Rv1258*, *Rv1634*, *Rv2688*, *Rv1273*, *Rv1819*, *Rv1458*, *Rv1877*, and *Rv1250* have been associated with drug resistance [23]. Most SNVs identified in EPs have been associated with an increase in drug tolerance. However, the converse may be true, such as found with loss of function mutations that prevent resistance, found naturally occurring in the *Rv1258* of Beijing lineage 2 isolates which inhibit RIF resistance [24]. Comprehensive genome-wide association studies between *M. tuberculosis* phenotypes and genotypes in the context of the contribution of EPs to drug resistance are not yet available. We studied phenotypic and genotypic profiles of 2221 *M. tuberculosis* genomes to investigate a possible role in the context of MDR, pre-extensively drug resistant (XDR), and XDR phenotypes. We used MTB-VCF (*M. tuberculosis*-Variant Call Format), an in-house bioinformatics pipeline to identify SNVs in EPs which were further analysed in the context of resistance genotypes.

2. Methods

2.1. Ethics statement

This study involved a review of databases and ethical approval was not required. The ReSeqTB database accessed was public, open, and anonymized with no link to patient data.

2.2. Analysis of *M. tuberculosis* genomes

We downloaded data of 2221 *M. tuberculosis* clinical isolates from the ReSeqTB database (Fig. 1). Genomes of *M. tuberculosis* isolates were downloaded using the Sequenced Read Archive (SRA) IDs provided in the ReSeqTB database linked with the National Center for Biotechnology Information (NCBI) website (SRA-NCBI) (<https://www.ncbi.nlm.nih.gov/sra>), Supplementary Table S1.

Phenotypic susceptibility profiles for these *M. tuberculosis* isolates were accessed through the DST profiles data in the ReSeqTB database (<https://platform.reseqtb.org/>). Details of *M. tuberculosis* specimens were entered as per their original genome submissions. Date of isolation were available for 521 of 789 isolates together with location data for 485 of them (Supplementary Table S2). These DR strains were collected between 1996 and 2013. The majority of samples were from Belarus (51.8%) and Russia (39.8%) with 5.4% from South Africa (5.4%).

2.3. Operational definitions of *M. tuberculosis* resistance phenotypes

We further used the following definitions to categorize resistance phenotypes for this study. MDR: RIF and INH-resistant *M. tuberculosis* isolates. Pre-XDR: MDR strains which are also resistant to a fluoroquinolone. XDR: MDR strains resistant to fluoroquinolone and an injectable aminoglycoside. The MDR, pre-XDR, and XDR definitions used in this analysis pre-date the WHO revised definitions for MDR-TB (2021) [25]. *M. tuberculosis* genomes were categorized using UNIX scripting based on available DST results into groups based on these operational definitions.

2.4. Bioinformatics analysis

FASTQ data of each *M. tuberculosis* isolate was used for bioinformatics analysis. These were first categorized based on their DST profiles available in ReSeqTB using UNIX scripting into four different data sets using the operational definitions used in this study: susceptible, MDR, pre-XDR, and XDR. Genome coordinates for 47 EP genes of interest were identified through Mycobrowser (<https://mycobrowser.epfl.ch/genes/>). Targeted variant calling was used to identify (single-nucleotide polymorphisms [SNPs] and INDELs) in each genome.

An in-house bioinformatics pipeline for targeted variant called “MTB-VCF” [26] described in Supplementary Fig. S1, was used to extract variants against the 47 EP genes using gene coordinates (start and stop position) extracted from NCBI gene database. Whole-genome analysis was carried out for each strain by comparing against the *M. tuberculosis* reference genome H37Rv NC_000962.3 [https://www.ncbi.nlm.nih.gov/genome/?term=NC_000962.3].

2.5. The MTB-VCF pipeline

Different tools were employed in our in-house MTB-VCF pipeline for calling single-nucleotide polymorphisms (Supplementary Fig. S1). The pipeline included BWA (<http://bio-bwa.sourceforge.net/>), SAMTOOLS (<http://samtools.sourceforge.net/>), Picard (<https://broadinstitute.github.io/picard/>), Genome Analysis Toolkit (GATK; <https://software.broadinstitute.org/gatk/>), and BCFTOOLS (<http://samtools.github.io/bcftools/bcftools.html>) for variant calling. These tools were run using default parameters. Variants and indels were called using the haplotype caller from GATK with a ploidy parameter of 1. Finally, all these tools were incorporated into a parallel script (<https://kb.parallels.com/121370>) to call variants within each data set all at once to generate a single multi-sample VCF file. The pipeline can be found on <https://github.com/safinaARK/MTB-VCF> (Razzak, 2023 #3496).

2.6. Variant calling

To investigate genetic variation in EP genes associated with drug resistance, we compared the reference H37Rv sequences of each EP gene in DR isolates with those of susceptible *M. tuberculosis* isolates in the study data set. This led to the identification of unique variants in the susceptible *M. tuberculosis* data set. Our goal

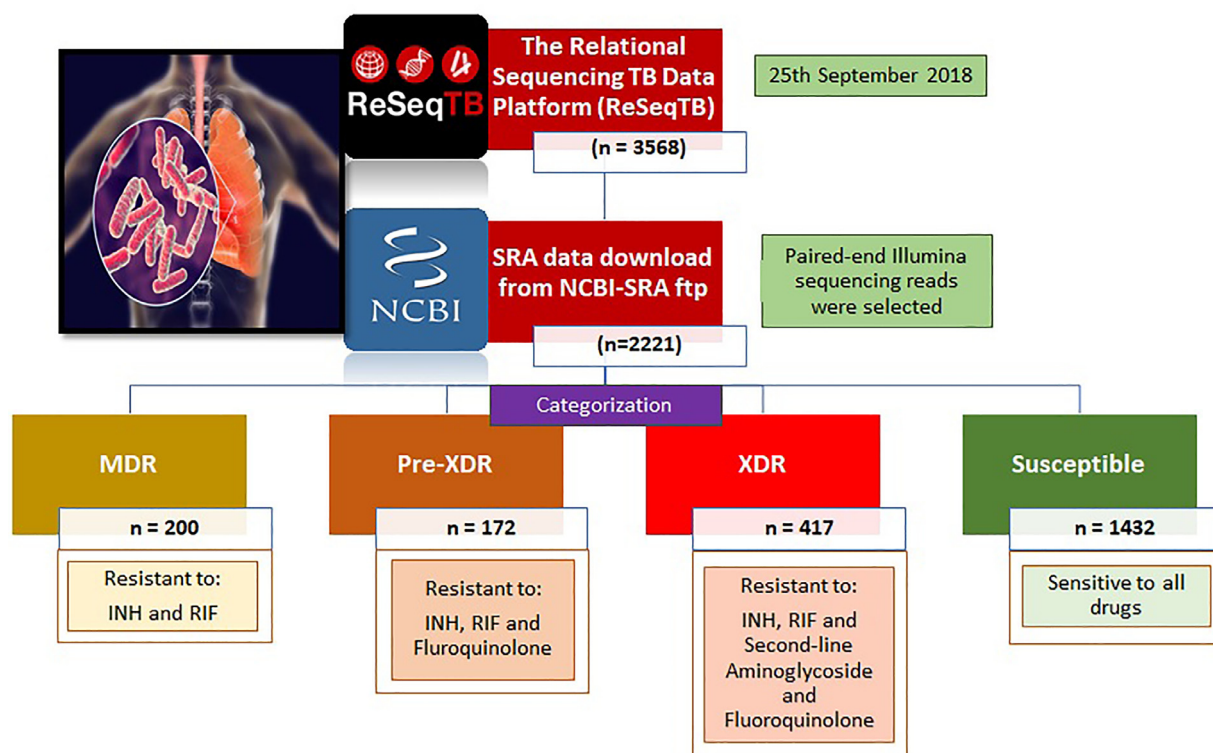


Fig. 1. ReSeqTB data curation and categorization. The flow chart depicts the manner in which genomic sequence data for *Mycobacterium tuberculosis* isolates were selected and downloaded from the ReSeqTB database and categorized based on their drug resistance phenotype prior to genetic analysis.

was to identify “unique” variants present in DR isolates. Therefore, comparisons were run between variants identified in found in susceptible *M. tuberculosis* genomes and those of DR isolates. Variants common to susceptible and DR groups were subtracted to allow identification of unique variants.

2.7. Phylogenogram

The fasta sequences of *M. tuberculosis* isolates divided into MDR, pre-XDR, and XDR data sets were aligned to generate multiple sequence alignment with MUSCLE (<https://www.ebi.ac.uk/Tools/msa/muscle/>). These were then bootstrapped to maximum likelihood phylogenetic trees of each data set aligned (.aln) file with RAxML (<https://cme.h-its.org/exelixis/web/software/raxml/>). The Newick files generated by RAxML were used to visualize trees in an online tree visualization tool called the Interactive Tree of Life (iTOL) (<https://itol.embl.de/>) along with annotation files that contain the genotypic variant information, and lineage information of each data set generated by TBProfiler [8].

2.8. Variant classification

SIFT/PolyPhen scoring (https://faculty.washington.edu/wjvs18/GS561/cSNPs_lab.html) was used to predict the impact of each variant as per a score of low, medium, or high to assess the significance of the variant on protein structure of the EP gene under study.

2.9. Statistical analysis

Nonparametric statistical analysis was performed using Mann-Whitney and Kruskal–Wallis tests to determine the correlation between lineages and variants using Prism (v5) (<https://www.graphpad.com/scientific-software/prism/>) software, $P < 0.05$ was considered significant.

3. Results

3.1. Selection of efflux pumps for investigation of drug resistance

We considered EPs which had been shown to be associated with drug resistance to key anti-TB drugs, RIF, INH, PZA, STR, EMB, BDQ, CFZ, cycloserine (CYC), ofloxacin (OFX), and ciprofloxacin (CIP). These EPs had been associated with drug resistance in *M. tuberculosis* due to gene expression studies or by causal association of their SNVs. The EPs included are listed in Table 1 according to their drug target and functional types. Nineteen EPs were MFS type transporters, Rv0037c, Rv0191, Rv0783c, Rv0849, Rv1250, Rv1258c, Rv1410c, Rv1634, Rv1877, Rv2333c, Rv2456c, Rv2459, Rv28994, Rv3239c, and Rv3728. Fourteen EPs were ABC type transporters; Rv0194, Rv1218c-Rv1217c, *rrrA* (Rv2936), *rrrB* (Rv2937), *rrrC* (Rv2938), Rv1273c-Rv1272c, and Rv2688c-Rv2687c-Rv2686c. Five were RND family proteins; Rv0206, Rv0507, Rv0676, Rv2942, and Rv3823, one was an SMR type (Rv3065), one was an APC type (Rv1704). Additionally, seven EPs were those identified as conserved hypothetical protein/ transcriptional regulator/ integral membrane protein based on their homology: Rv0678, Rv0842, Rv1257, Rv2209, Rv3000, Rv3008, and Rv3197.

3.2. Selection and categorization of *M. tuberculosis* isolates

ReSeqTB classifies strains as MDR if they have phenotypic resistance to both RIF and INH. Our primary data set contained 1432 susceptible and 789 MDR isolates. We used UNIX scripting based on our operational definition of pre-XDR and XDR to identify that the DR genomes we had downloaded comprised 200 MDR, 172 pre-XDR, and 417 XDR isolates.

We observed that 9% belonged to lineage 1, 28.77% to lineage 2, 13.28% to lineage 3, 42.63% to lineage 4, 5.49% to lineage 5, and 0.81% were lineage 6 isolates (Table 2). Lineage 4 isolates were

Table 1
Efflux pump genes investigated for genotypic variation in *Mycobacterium tuberculosis* isolates.

Efflux pump	Anti-tuberculous drug	Type of EP	References
Rv0191	Pyrazinamide	Major facilitator superfamily (MFS)	[27]
Rv0194	Isoniazid, streptomycin, and ethambutol	ATP-binding transport protein family (ABC transporters)	[31,42]
Rv0206c (mmpL3)	SQ109	RND superfamily	[43]
Rv0450c (mmpL4)	Rifampicin	ATP-binding transport protein family (ABC transporters)	[33]
Rv0507 (mmpL2)	Isoniazid	RND superfamily	[42]
Rv0676c (mmpL5)	Azoles, bedaquiline, and rifampicin	RND superfamily	[33,44]
Rv0678	Azoles, clofazimine, and bedaquiline	Conserved protein	[33,44]
Rv0783c (emrB)	Rifampicin	Major facilitator superfamily (MFS)	[45]
Rv0842	Isoniazid	Probable conserved integral membrane protein	[42]
Rv0849	Isoniazid	Major facilitator superfamily (MFS)	[42,46]
Rv1218c-Rv1217c	Isoniazid and rifampicin	ATP-binding transport protein family (ABC transporters)	[47]
Rv1250	Isoniazid	Major facilitator superfamily (MFS)	[42]
Rv1257c	Streptomycin	Probable oxidoreductase	[48]
Rv1258c	Isoniazid, tetracycline, rifampicin, and clofazimine	Major facilitator superfamily (MFS)	[49]
Rv1272c-1273c	Isoniazid	ATP-binding transport protein family (ABC transporters)	[42]
Rv1410c (p55)	Isoniazid, tetracycline, rifampicin, and clofazimine	Major facilitator superfamily (MFS)	[42,46]
Rv1458c, Rv1457c, Rv1456c	Isoniazid, rifampicin, ethambutol, and streptomycin	ATP-binding transport protein family (ABC transporters)	[28,42]
Rv1634	Isoniazid	Major facilitator superfamily (MFS)	[42]
Rv1667c	Pyrazinamide	ATP-binding transport protein family (ABC transporters)	[27]
Rv1686c-Rv1687c	Isoniazid	ATP-binding transport protein family (ABC transporters)	[42]
Rv1704c (cycA)	Cycloserine	Amino acid permease family (APC family)	[50]
Rv1877	Streptomycin	Major facilitator superfamily (MFS)	[31]
Rv2209	Ofloxacin	Probable conserved integral membrane protein	[21]
Rv2333c (stp)	Isoniazid, spectinomycin, and tetracycline	Major facilitator superfamily (MFS)	[51]
Rv2459 (jefA)	Isoniazid and ethambutol	Major facilitator superfamily (MFS)	[21,46,52]
Rv2477c	Ofloxacin	ATP-binding transport protein family (ABC transporters)	[21]
Rv2686c, Rv2687c, Rv2688c	Streptomycin and ciprofloxacin	ATP-binding transport protein family (ABC transporters)	[21,36]
Rv2846c (efpA)	Isoniazid	Major facilitator superfamily (MFS)	[21,42]
Rv2937 (drrB)	Isoniazid	ATP-binding transport protein family (ABC transporters)	[46]
Rv2942(mmpL7)	Isoniazid	RND superfamily	[18,53]
Rv2994	Isoniazid and rifampicin	Major facilitator superfamily (MFS)	[18,21]
Rv3000	Bedaquiline	Possible conserved transmembrane protein	[54]
Rv3008	Pyrazinamide	Hypothetical protein	[27]
Rv3065 (Mmr)	Isoniazid and ethambutol	Small multidrug resistance (SMR) protein family	[18,21]
Rv3197A (whiB7)	Rifampicin	Probable transcriptional regulatory protein WhiB-like WhiB7	[55]
Rv3239c	Rifampicin	Major facilitator superfamily (MFS)	[55]
Rv3728	Isoniazid and ethambutol	Major facilitator superfamily (MFS)	[21]
Rv3756c (proZ)	Pyrazinamide	ATP-binding transport protein family (ABC transporters)	[27]
Rv3823c (mmpL8)	Isoniazid	RND superfamily	[42]

Table 2
Phylogenetic and phenotypic description of *M. tuberculosis* isolates.

MTB lineage	Isolates per lineage (%)	Susc. (%)	MDR (%)	pre-XDR (%)	XDR (%)
Isolates per dataset	2221	1432	200	172	417
1	222 (9.99)	187 (13.05)	11 (5.50)	16 (9.30)	8 (1.91)
2	592 (26.66)	224 (15.64)	103 (51.50)	91 (52.90)	174 (41.72)
3	289 (13.01)	230 (16.06)	10 (5.0)	3 (1.74)	46 (11.03)
4	1110 (49.97)	784 (54.74)	75 (37.50)	62 (36.04)	189 (45.32)
5	1 (0.04)	0	1 (0.50)	0	0
6	7 (0.30)	7 (0.48)	0	0	0

Susc., susceptible; MDR, multidrug resistant; pre-XDR, MDR strains with fluoroquinolone resistance; XDR, MDR strains resistant to fluoroquinolone and at least one injectable aminoglycoside. Whole-genome sequences of MTB isolates were downloaded from ReSeqTB database (<https://platform.reseqtb.org/>) based on their phenotypic drug sensitivity testing (DST) profiles.

predominant amongst all isolates (49.97%) and susceptible isolates (54.7%). However, amongst DR isolates, lineage 2 were found to be predominant comprising, MDR (51.5%) and pre-XDR (52.9%) isolates, with 41.72% of XDR isolates.

3.3. Efflux pump gene variants found in drug-resistant *M. tuberculosis*

The MTB-VCF pipeline was used to call variants in the 47 EP genes of interest by comparison with the *M. tuberculosis* H37Rv reference genome. This was done separately for each data set; susceptible, MDR, pre-XDR, and XDR. We observed 1436 SNVs in drug susceptible isolates which were not present in the H37Rv reference strain. These variants comprised non-coding, synonymous,

frameshift, missense, stop gain, stop loss, and start loss variants, as listed in Supplementary Table S3. Variants common to susceptible and DR groups were “subtracted,” leaving behind only “unique” variants, each in MDR, pre-XDR, and XDR isolates. Analysis of 47 EP genes amongst DR isolates identified 418 SNVs amongst DR isolates, of which 91 (53.5%) were found in MDR isolates, whereas 153 (68.9%) were associated with pre-XDR and 174 (61.3%) with XDR isolate genomes.

3.4. High-impact efflux pump gene variants and their frequency in DR genomes

In total, 231 unique SNVs were identified in DR genomes (MDR, pre-XDR, and XDR) and were found across 20 different EPs.

SIFT/PolyPhen analysis categorises the variants into low ($n = 147$), medium ($n = 46$), or high ($n = 38$) impact types (Supplementary Table S4). As high-impact SNVs are most likely to result in a functional effect on proteins, we focused on these. As shown in Table 3, there were 23 previously reported unique high-impact SNVs and 15 novel variants. Their occurrence varied between EPs; Rv0194 ($n = 8$); Rv0676, Rv1819, and Rv1877 ($n = 3$, each respectively); Rv0450, Rv1217, Rv1634, Rv1704, and Rv2688 ($n = 2$, each respectively). We found a single SNV each for the remaining 11 EP. We investigated the occurrence of these EP SNVs in the MDR/pre-XDR/XDR data sets. SNVs which occurred at the greatest frequency of $\geq 50\%$ within any of the three DR subsets were found in: Rv0191, Rv0194, Rv1217, Rv1250, Rv0676, Rv1704, Rv1877, and Rv3756. SNVs at a frequency of 49% to 20% were found in: Rv0194, Rv0450, Rv0507, Rv0676, Rv1218, Rv1273, Rv1458, Rv1634, Rv1704, Rv1819, Rv1877, Rv2688, Rv3728, and Rv3823 (Table 3). An additional 11 SNVs in EPs; Rv0194, Rv0450, Rv0507, Rv0676, Rv1218, Rv1273, Rv1458, Rv1634, Rv1704, Rv1819, Rv1877, Rv2688, Rv3728 and Rv3823, occurred at frequencies $< 20\%$. The least frequent SNVs were Rv0194 fs915 and 695Q>*, present in a single isolate each.

To investigate the association between the abovementioned SNVs and drug resistance in *M. tuberculosis*, we compared their presence with pre-XDR, MDR, and XDR subgroups using Chi-squared analysis. Sixteen high-impact SNVs found across 12 EP genes were found to be significantly associated with DR genomes ($P \leq 0.05$; Table 3). These were: Rv0191 213A>T, Rv0507 426R>H, Rv0676c 948I>V, and 794T>I, Rv1217 173A>T, Rv1218 243Q>R, Rv1273 462G>K, Rv1458 133T>A, Rv1819 551sg, and 273I>T, Rv1877 370W>G, and 591 Y>sg, Rv2333 7771fs, Rv2688 213C>R, and 156P>T, and Rv3728 325W>sg. Of these, 12 had been reported previously [27–29]. The four SNVs novel from this study; Rv1877 370W>G and 591>*, Rv2333 771#, and Rv3728 325W>*, respectively, were enriched amongst pre-XDR and XDR isolates.

3.5. Frequency of high-impact unique SNV in different lineages

Polymorphisms associated with *M. tuberculosis* lineages may have a confounding effect on the curation of variants associated with drug resistance. Therefore, we investigated whether the 38 SNVs in EP genes could be associated with the *M. tuberculosis* lineage of the strain in which they occurred. We found that the unique SNVs EP were not associated with any particular *M. tuberculosis* lineages (Table 4).

3.6. Understanding the context of high-impact SNVs in EP genes in phenotype-genotype associations

We hypothesized that variants in EP genes could enhance drug resistance in *M. tuberculosis* isolates in the absence of a known resistance genotype. For this, we considered the DR genotype of the 789 *M. tuberculosis* isolates studied and identified those with phenotype-genotype discrepancies. Phenotypes were as per the Re-SeqTB DST information provided. The DR genotype of the isolates was identified using the bioinformatics pipeline, TB-Profiler. The TB-Profiler predicts genotypic resistance to RIF and INH with *rpoB*, *katG*, and *inhA* (promoter) regions. It predicts genotypic resistance to FLQs with *gyrA* and *gyrB* mutations, and aminoglycoside (AMN) resistance with *rrs* mutations. Of the 789 *M. tuberculosis* genomes analysed using the TB-Profiler, there were 52 isolates with phenotype-genotype discrepancies. Twenty-nine were MDR, 11 were pre-XDR isolates, and 13 were XDR strains. Notably, none of the 52 DR isolates had an *rpoB* mutation, which is most often associated with RIF resistance. Forty-six of 52 isolates did not have either *katG* or *inhA* mutations, usually associated with INH resistance. None of the pre-XDR or XDR strains had *gyrA* or *gyrB* mutations which are most often associated with FLQ resistance. In 9

of 13 XDR strains, *rrs* mutations, often associated with aminoglycoside resistance were absent (Supplementary Table S5).

The pattern of 16 high-impact SNVs in EPs found associated with drug resistance within the 52 discrepant isolates is depicted in Fig. 2. Each of the 52 discrepant genomes had at least one high-impact SNV present. The SNVs occurred variably as shown in Fig. 2, with frequencies of 71.4% for Rv0191 213A>T; 30.8% for Rv0507 426R>H; Rv0676c 948I>V, 36.5% and 794T>I, 21.1%; Rv1217 173A>T, 32.7%; Rv1218 243Q>R, 15.8%; Rv1273 462G>K, 21.2%; Rv1458 133T>A, 21.2%; Rv1819 551-fs, 11.5% and 273I>T, 25%; Rv1877 370W>G, 25% and 591 Y>sg, 36.5%; Rv2333 7771fs, 7.8%; Rv2688 213C>R, 9.6% and 156P>T, 25%; and Rv3728 325W>sg, 23.1%. There was more SNVs present in pre-XDR and XDR as compared with MDR genomes, Kruskal-Wallis analysis, $P < 0.0001$. Interestingly, Rv0191_213A>213T was present in all 28 MDR and 10 of 11 pre-XDR strains identified to have discrepant genotypes but was not found in any of the 13 XDR strains. Twenty of 28 MDR genomes only had Rv0191 213>T. Most discrepant XDR genomes had mutations in multiple EPs, with between 2 and 11 SNVs occurring in between 9 and 10 genes within each isolate. Notably, SNVs which occurred in 92% to 100% of the XDR isolates were: Rv0676 173A>T, Rv1217 173 T > A, Rv1817 273 I > T, Rv1877 591Y>sg, and Rv2866 156 P > T, respectively.

3.7. Drug resistance effect of EP genes

The 52 DR isolates we identified had phenotype-genotype discrepancies related to RIF, INH, FLQ, and AMN resistance. We tried to link the 16 high-impact SNVs present in 12 EP genes in the context of the observed discrepancies between phenotypic and genotypic drug resistance. RIF resistance has been associated with Rv0676, Rv1217_1218, and Rv1819 (Table 5). INH resistance has been associated with; Rv0507, Rv1217_1218, Rv1273, Rv1458, Rv2333, and Rv3728. FLQ resistance has been associated with Rv2688.

4. Discussion

Through the genome analysis of 2221 drug-susceptible and 789 drug-resistant *M. tuberculosis* isolates, we identified variants in EP genes associated with resistance. Thirty-eight high-impact SNVs were found across 20 EP genes associated with resistance to RIF, INH, FLQ, and aminoglycosides. The SNVs included 25 previously reported and 13 novel variants and were found in MDR, pre-XDR, and XDR isolates with phenotype-genotype discrepancies where mutations in the known resistance-conferring mutations are found to be absent. Sixteen of the high-impact SNVs found across 12 EP genes were significantly associated with drug resistance. Twelve of these SNVs were found across 10 EP genes in strains which showed phenotype-genotype discordance. These SNVs occurred significantly more in XDR as compared with MDR isolates.

There is an increasing appreciation of the role of genomics in the prediction of drug resistance in *M. tuberculosis*. WGS has become increasingly more available due to expansion in technologies, access to equipment, and establishment of stable wet-lab and bioinformatics pipelines. There are many analysis platforms that are open source and provide guidance on interpretation of genome data. However, accurate prediction of drug resistance requires comprehensive information on the role of genes involved in conferring drug resistance, the mechanism of drug and target interactions, and the role of additional mechanisms that may enhance the drug effect. EPs can contribute to drug resistance by exporting a range of biological molecules across the cell membrane. They may act in a nonspecific manner with a single pump transporting out multiple drugs. *In vitro* and *in vivo* studies have identified the modification of gene expression of EPs in response to drug pressure. The challenge is to have a method that may predict the effect of EPs on

Table 3

Genetic variants in efflux pump genes with high SIFT/PolyPhen scores associated with drug-resistant MTB strains.

Efflux pump	Chr POS	REF (H37Rv)	ALT	Effect	Protein Change	Datasets	No. isolates	% MDR (n = 200)	% pre-XDR (n = 172)	% XDR (n = 417)	p-value	Reported/ Novel
Rv0191	222,925	G	A	Missense	213A>213T	PreXDR/MDR	111/191	95.5	64.5		[^] 0.000	[27]
Rv0194	227,098	T	C	Missense	74M>74T	PreXDR	121		70.3		–	[29]
Rv0194	227,685	A	ATTACGGTGGG CACCTTTGTTGCC	Frameshift	277#	MDR	15	7.5			–	Novel
Rv0194	227,878	TGGAG TTCCA	T	frameshift	334LEFQ>334L	PreXDR	77		44.8		–	novel
Rv0194	228,506	C	CG	stop_gain	682Q>682*	MDR	11	5.5			–	novel
Rv0194	228,960	C	T	stop_gain	695Q>695*	PreXDR	1			0.2	–	novel
Rv0194	229,401	GT	G	frameshift	842#	XDR	15	6.0		3.8	–	novel
Rv0194	230,170	C	A/T	missense	1098P>1098L	MDR/XDR	12/16				[^] 0.227	[29]
Rv0194	229,621	C	CTGCT	frameshift	915#	PreXDR	1		0.6		–	novel
Rv0450c	539,120	C	A	missense	791R>791L	PreXDR	66		35.5		–	novel
Rv0450c	539,500	C	G	missense	664M>664I	PreXDR	61		44.2		–	novel
Rv0507	598,475	G	A	missense	426R>426H	PreXDR/XDR	76/111		44.2	26.6	[^] 0.000	[28]
Rv0676c	775,639	T	C	missense	948I>948V	PreXdr/XDR	29/321		16.9	77.0	[^] 0.000	[28]
Rv0676c	776,100	G	A	missense	794T>794I	PreXdr/XDR	24/121		14.0	29.0	[^] 0.000	[28]
Rv0676c	777,451	C	G	missense	344V>344L	XDR	12			2.9	–	[56]
Rv1217c	1,361,285	C	T	missense	173A>173T	PreXDR/XDR	89/234		51.7	56.1	[^] 0.000	[29]
Rv1217c	1,361,350	A	C	missense	151L>151R	XDR	301			72.2	–	[29]
Rv1218	1,362,006	T	C	missense	243Q>243R	MDR/PreXDR/XDR	1/112/12	0.5	65.1	2.9	^{^^} 0.000	[29]
Rv1250	1,395,010	A	G	missense	278R>278G	XDR	312			74.8	–	[34]
Rv1273c	1,422,667	C	T	missense	462G>462K	PreXDR/XDR	71/121		41.3	29.0	[^] 0.004	[29]
Rv1458c	1,643,864	T	C	missense	133T>133A	PreXDR/XDR	32/130		18.6	31.2	[^] 0.002	[29,31]
Rv1634	1,839,306	A	G	missense	47I>47V	XDR	123			38.3	–	[29]
Rv1634	1,839,759	G	C	missense	198G>198R	PreXDR/XDR	11/23		6.4	5.5	0.677	[29]
Rv1704c	1,929,979	G	A	missense	493A>493V	PreXDR	111		64.5		–	novel
Rv1704c	1,931,179	C	A	missense	93R>93L	XDR	123			29.5	–	[57]
Rv1819c	2,063,078	G	GC	frameshift	551#	MDR/PreXDR/XDR	32/21/11	16.0	12.2	2.6	^{^^} 0.000	[31]
Rv1819c	2,063,911	A	G	missense	273I>273T	PreXDR/XDR	18/291		10.5	69.8	^{^^} 0.000	[29]
Rv1819c	2,062,922	T	C	missense	603I>603V	XDR	112			26.9	–	novel
Rv1877	2,127,011	T	G	missense	370W>370G	PreXDR/XDR	49/154		28.5	36.9	[^] 0.05	novel
Rv1877	2,127,504		C	missense	534I>534T	XDR	12			2.9	–	[29]
Rv1877	2,127,676	T	G	stop_gain	591Y>591*	PreXDR/XDR	111/211		64.5	29.0	[^] 0.000	novel
Rv2333c	2,608,091	T	TA	frameshift	77I#	PreXDR/XDR	32/27		18.6	6.5	[^] 0.000	novel
Rv2688c	3,005,014	A	G	missense	213C>213R	PreXDR/XDR	77/11		44.8	2.6	[^] 0.000	[29]
Rv2688c	3,005,185	G	T	missense	156P>156T	PreXDR/XDR	12/198		7.0	47.5	[^] 0.000	[29]
Rv3008	3,367,041	T	TGAA	frameshift	133L>133LN	MDR	23	11.5			–	novel
Rv3728	4,175,847	G	A	stop_gain	325W>325*	PreXDR/XDR	49/154		28.5	36.9	[^] 0.05	novel
Rv3756c	4,202,052	CGAT	C	inframe_deletion	187IA>187A	PreXDR	123		71.5		–	[27]
Rv3823c	4,288,284	C	T	stop_gain	1082W>1082*	XDR	145			34.8		[27]

“*” stop codon, “#” change of greater than 10 amino acids.

Pearson's chi squared test, “^,” 2 × 2 table; “^^,” 3 × 2 table was used to calculate P values between groups where SNV occurred in more than one DR sub-group.

Table 4

Lineage-wise occurrence of SNVs in efflux pump genes of MDR TB isolates.

		Lineage 1	Lineage 2	Lineage 3	Lineage 4	Lineage 5	Lineage 6
Rv0191	90V>90L	–	20	2	2	–	–
	179V>179A	2	19	21	4	–	–
	213A>213T	4	198	16	91	–	1
	339A>339V	–	9	–	2	–	–
	344G>344D	–	21	4	8	–	–
	410P>410L	2	16	13	2	–	–
Rv0194	74M>74T	–	94	1	26	–	–
	114G>114S	3	15	21	4	–	–
	156V>156A	–	11	1	9	–	–
	277-fs	–	4	–	11	–	–
	334LEFQ>334L	–	55	1	21	–	–
	528R>528C	1	19	16	8	–	–
	542D>542E	–	5	1	6	–	–
	682Q>682*	–	2	–	10	–	–
	695Q>695*	–	–	–	1	–	–
	853P>853L	–	8	–	5	–	–
	842-fs	–	13	1	1	–	–
	1098P>1098L	–	20	4	4	–	–
	915-fs	–	–	–	1	–	–
	791R>791L	–	46	1	19	–	–
	664M>664I	–	43	1	17	–	–
Rv0450c	568D>568Y	–	16	1	14	–	–
	426R>426H	5	165	29	40	–	–
	435E>435Q	–	2	–	10	–	–
Rv0507	464S>464G	–	2	–	11	–	–
	734#	–	–	–	1	–	–
	948I>948V	5	221	29	89	–	–
Rv0676c	794T>794I	5	73	29	34	–	–
	946R>946*	–	3	–	–	–	–
	344V>344L	–	11	–	1	–	–
Rv0678	39#	–	1	–	–	–	–
	64#	–	–	–	1	–	–
	323Y>323*	–	1	–	–	–	–
Rv0783c	77Y>77*	–	1	–	–	–	–
Rv0849	196#	–	5	–	–	–	–
	276#	–	1	–	–	–	–
	4#	–	1	–	–	–	–
Rv1217c	173A>173T	5	212	28	72	–	–
	151L>151R	5	188	28	74	–	–
	243Q>243R	–	102	3	20	–	–
Rv1250	278R>278G	5	199	29	76	–	–
Rv1258c	247#	–	1	–	–	–	–
Rv1273c	462G>462K	5	121	28	34	–	–
Rv1458c	133T>133A	5	93	28	32	–	–
Rv1557	158A>158G	–	14	–	4	–	–
	191#	–	1	–	–	–	–
	343#	–	1	–	–	–	–
Rv1634	47I>47V	5	61	28	26	–	–
	137P>137L	–	29	–	5	–	–
	198G>198R	2	6	18	7	–	–
Rv1704c	493A>493V	–	91	–	20	–	–
	290L>290V	–	4	1	6	–	–
	188P>188A	2	2	11	–	–	–
Rv1819c	93R>93L	5	60	28	26	–	–
	551#	1	32	9	23	–	–
	273I>273T	5	193	28	77	–	–
Rv1877	603I>603V	5	51	28	24	–	–
	59G>59D	–	26	–	6	–	–
	370W>370G	5	119	27	48	–	–
Rv2333c	534I>534T	2	–	10	–	–	–
	591Y>591*	5	282	26	78	–	–
	237C>237*	–	1	–	–	–	–
Rv2459	139#	–	1	–	–	–	–
	77I#	5	23	18	13	–	–
	483#	1	–	4	–	–	–
Rv2686c	59#	1	–	4	–	–	–
Rv2688c	213C>213R	–	72	–	16	–	–
	156P>156T	5	120	22	56	–	–
	33#	–	1	–	–	–	–
Rv2942	185R>185H	–	43	–	12	–	–
Rv3000	209LVSLPAATRR*	–	1	–	–	–	–
Rv3008	33W>33*	–	91	–	20	–	–
	133L>133LN	1	–	9	13	–	–
	134#	–	1	–	10	–	–

(continued on next page)

Table 4 (continued)

		Lineage 1	Lineage 2	Lineage 3	Lineage 4	Lineage 5	Lineage 6
Rv3197	84GRPRK>84K	–	1	–	–	–	–
	12#	–	1	–	–	–	–
Rv3239c	826#	–	1	–	–	–	–
	508#	–	1	–	–	–	–
	426V>426AVLGV	–	–	2	–	–	–
	164#	–	1	–	–	–	–
Rv3728	325W>325*	5	121	28	44	–	–
	475VGI>475V	–	1	–	–	–	–
Rv3756c	1871A>187A	–	96	1	27	–	–
	171#	–	1	–	–	–	–
	35A>35E	–	5	1	6	–	–
Rv3823c	1082W>1082*	5	79	28	29	–	–
	161#	–	4	–	–	–	–

Table 5
High-impact efflux pumps SNV and their association with discrepant drug resistance MTB isolates.

Anti-tuberculous drug	Efflux pump	SNVs
Bedaquiline and rifampicin	Rv0676c	794T>I, 948I>948V
Isoniazid and rifampicin	Rv1217	173A>173T
Isoniazid and rifampicin	Rv1218	243Q>243R
Isoniazid and rifampicin	Rv1819c	273I>T, 551#
Streptomycin	Rv1877	370W>370G
Streptomycin and ciprofloxacin	Rv2688c	156P>T, 213C>213R
Pyrazinamide	Rv0191	213A>213T
Isoniazid	Rv0507	426R>426H
Isoniazid	Rv2333c	771#
Isoniazid and ethambutol	Rv3728	325W>325*
Isoniazid	Rv1273c	462G>462K
Isoniazid, rifampicin, ethambutol, and streptomycin	Rv1458c	133T>133A

The table shows the occurrence of 16 high-impact SNVs in EP genes which were identified amongst 52 MDR TB isolates with discrepant phenotype-genotype correlations. EP, efflux pump; MDR, multidrug resistance; SNV, single-nucleotide variants.

a specific drug without complex experimentation. Our study comprised analysis of DR isolates from the ReSeqTB database where strains were listed as per their DST phenotypes. Although there have been many studies that have identified the presence of such SNVs in EP genes, few of them have used in silico prediction to determine their impact on the gene function.

We used MTB-VCF, an in-house automated variant calling pipeline tool to call SNVs in EPs [30]. These were further categorized based on their SIFT/PolyPhen scores and high-impact SNVs were identified. Phenotype-genotype correlations in *M. tuberculosis* isolates were determined using the DST profiles available in ReSeqTB compared with the resistance genotype determined by the TB-Profiler [8]. The identification of 52 discrepant isolates highlights the complexity of using whole-genome data for prediction of drug resistance phenotypes.

Strains with discordance in RIF resistance did not have any *rpoB* gene mutations. We found that phenotypically RIF-resistant strains without *rpoB* mutations had SNVs in *Rv0194*, *Rv1217-Rv1218*, *Rv1819*, and *Rv0450*. *Rv0194* 74M>T and 1098P>L have been associated with INH, STR, and EMB resistance [29,31]. *Rv1217-Rv1218* is associated with INH and RIF resistance [29] and we observed high-impact mutations: 173A>T in 55%, 151L>R at 51%, and 243R>R at 21% in pre-XDR/XDR strains. *Rv1819* has been associated with resistance to RIF and INH [32]; we observed three high-impact mutations, fs at c.551, 273I>T, and 603I>V of which two have been reported in literature [29,31]. *Rv0450* has previously been associated with RIF resistance [33]. This is the first study to identify genetic variants in *Rv0450*, whereby we observed 791R>L and 664M>I to be present in 39% and 36% of pre-XDR isolates, respectively.

Rv1458 133T>A has been associated with resistance to INH, RIF, EMB, and STR [31], and we found it to be present in 28% of pre-XDR and XDR strains. *Rv0507* 426R>I is associated with INH resistance [28], we found this mutation in 27% XDR and 44% pre-XDR

strains. *Rv1250* 278R>G has been associated with INH resistance [29,34], here, we found it in 53% of XDR strains.

Pre-XDR and XDR strains had a larger number of SNVs than MDR strains, with SNVs across up to 17 different EP genes. Discrepant pre-XDR and XDR strains (FLQ and aminoglycoside resistance without *gyrA*, *gyrB*, or *rrs* mutations) displayed SNVs in *Rv1634* and *Rv2686*. *Rv1634* is reported to be involved in [norfloxacin](#) and [ciprofloxacin](#) efflux [35]. We observed *Rv1634* mutations 471>V and 98G>R [29], to be associated with 29% XDR and 6% pre-XDR and XDR strains. Upregulation of *Rv2688c* has been associated with FLQ resistance, *Rv2688c* along with *Rv2686c*, and *Rv2687c* actively pumps out ciprofloxacin [36]. We found variants *Rv2688c* 213C>R and 156P>T [29], to be present in 15% and 36% of pre-XDR/XDR strains.

A limitation of this study is that it is based on genomic analysis of MDR TB strains available in the ReSeqTB database and full information was not available regarding the period of collection and origin of the isolates. Based on what was available (486/789 isolates), the majority of submissions (91%) of isolates were from Belarus and Russia. Therefore, we are unable to investigate region-wise association of these isolates. At the time we initiated the study, the ReSeqTB database did not have any named *M. tuberculosis* genome submission from Pakistan. However, we subsequently examined *M. tuberculosis* DR genome data for SNVs in EP genes. In MDR-TB strains from Pakistan, we have observed *Rv0676* (mmpI5) mutations to be present in strains resistant to bedaquiline [37].

Our data further expands on the need to consider SNVs in EP genes as part of whole-genome data evaluation of *M. tuberculosis* strains. A recent report of an outbreak clone missed by Xpert has identified high rates of bedaquiline resistance [38]. Resistance to first-line drugs due to unique *M. tuberculosis* SNVs may have a compounding effect for the development of additional resistance [39]. Further understanding the association between EP SNVs and

ID		ReSeqTB-DST	Rv0191	Rv0507	Rv0676		Rv1217	Rv1218	Rv1273	Rv1458	Rv1819		Rv1877		Rv2333	Rv2688		Rv3728
		Class	213A>T	426R>H	948I>V	794T>I	173A>T	243Q>R	462G>K	133T>A	551#	273I>T	370W>G	591Y>*	77I#	213C>R	156P>T	325W>*
		(n)	38	16	19	11	17	8	11	11	6	13	13	19	4	5	13	12
S1	ERR04 0119	MDR	Y								Y							
S2	ERR04 0133	MDR	Y								Y							
S3	ERR06 7579	MDR	Y								Y							
S4	ERR06 7591	MDR	Y								Y							
S5	ERR06 7595	MDR	Y								Y							
S6	ERR06 7597	MDR	Y								Y							
S7	ERR06 7600	MDR	Y															
S8	ERR06 7602	MDR	Y															
S9	ERR06 7675	MDR	Y															
S10	ERR06 7678	MDR	Y															
S11	ERR06 7688	MDR	Y															
S12	ERR06 7726	MDR	Y															
S13	ERR06 7734	MDR	Y															
S14	ERR06 7740	MDR	Y															
S15	ERR13 3801	MDR	Y															
S16	ERR13 3803	MDR	Y															
S17	ERR13 3823	MDR	Y															
S18	ERR13 3824	MDR	Y															
S19	ERR13 3842	MDR	Y															
S20	ERR13 3921	MDR	Y															
S21	ERR13 7215	MDR	Y															
S22	ERR22 7994	MDR	Y															
S23	ERR22 8016	MDR	Y															
S24	ERR22 8038	MDR	Y															
S25	ERR23 4557	MDR	Y															
S26	ERR23 4616	MDR	Y															
S27	ERR23 4645	MDR	Y					Y										
S28	SRR21 01753	MDR	Y												Y			
S29	ERR06 7614	pre-XDR	Y	Y	Y	Y									Y			
S30	ERR06 7606	pre-XDR	Y	Y	Y	Y									Y			
S31	ERR06 7607	pre-XDR	Y	Y	Y	Y									Y			
S32	ERR06 7617	pre-XDR	Y	Y	Y	Y												
S33	ERR06 7684	pre-XDR	Y	Y	Y			Y					Y	Y				

Fig. 2. Genotype-phenotype association of DR genomes and efflux pump (EP) gene variants. The heatmap depicts the occurrence of 16 high-impact single-nucleotide variants (SNVs) in individual EPs (shaded cell) across the 52 *Mycobacterium tuberculosis* genomes where a discrepancy between drug susceptibility testing (DST) phenotypes and genotypes (as per TB Profiler) were identified. Y-axis represents the resistant isolate. X-axis represents the SNV in each gene. SNVs were called using MTB-VCF tool. **n** in each RvXXX gene represents the number of SNVs in each gene.

S34	ERR06 7730	pre-XDR	Y	Y	Y			Y					Y	Y		Y	Y	Y
S35	ERR11 7465	pre-XDR	Y	Y			Y	Y	Y	Y			Y	Y		Y		Y
S36	ERR13 3798	pre-XDR	Y	Y			Y	Y	Y	Y			Y	Y		Y		Y
S37	ERR13 7263	pre-XDR	Y	Y			Y	Y	Y	Y			Y	Y		Y		Y
S38	ERR14 4587	pre-XDR	Y	Y			Y	Y	Y	Y	Y			Y		Y		
S39	SRR67 1790	pre-XDR					Y	Y						Y				
S40	SRR11 59066	XDR		Y	Y	Y	Y		Y	Y		Y	Y	Y			Y	Y
S41	SRR11 59286	XDR		Y	Y	Y	Y		Y	Y		Y	Y	Y			Y	Y
S42	SRR11 59926	XDR		Y	Y	Y	Y		Y	Y		Y	Y	Y			Y	Y
S43	SRR11 59957	XDR		Y	Y	Y	Y		Y	Y		Y	Y	Y			Y	Y
S44	SRR11 62517	XDR		Y	Y	Y	Y		Y	Y		Y	Y	Y			Y	Y
S45	SRR11 62530	XDR		Y	Y	Y	Y		Y	Y		Y	Y	Y			Y	Y
S46	SRR11 63096	XDR			Y	Y	Y		Y	Y		Y	Y	Y			Y	Y
S47	SRR11 63347	XDR			Y		Y					Y	Y	Y			Y	Y
S48	SRR11 69054	XDR			Y		Y					Y		Y			Y	
S49	SRR11 69118	XDR			Y		Y					Y		Y			Y	
S50	SRR11 72155	XDR			Y		Y					Y		Y			Y	
S51	SRR11 72184	XDR			Y		Y					Y		Y			Y	
S52	SRR67 1782	XDR			Y							Y						

Fig. 2. Continued

may help explain discrepancies in the identification of resistance, such as for rifampicin where *rpoB* mutations are absent, observed in India [40], or epidemic strains in Swaziland [41] and Eswantini [38]. Further investigations regarding these SNVs are required to correlate them using in vitro functional studies.

5. Conclusions

This is the largest data set of *M. tuberculosis* drug-resistant strain genomes analysed for mutations in EP genes. SNVs in EPs have the potential to contribute to either drug resistance or tolerance. Therefore, it may be valuable to include them into *M. tuberculosis* genome screening algorithms that identify high-risk SNVs for drug resistance genotyping.

Competing interests

The authors have no competing interests to declare.

Consent to participate

Not applicable

Availability of data and materials

Data are available in the supplementary files provided with this manuscript.

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Authors' contributions

ZH and RH conceived and supervised the study. SAR conducted the analysis. SAR and AK interpreted the data. ZH, SAR, AK, SS, and RH wrote the manuscript.

Ethics approval

Not required.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2024.05.006](https://doi.org/10.1016/j.jgar.2024.05.006).

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